

Question description

A game consists of n rooms and m teleporters. At the beginning of each day, you start in room 1 and you have to reach room n . You can use each teleporter at most once during the game. How many days can you play if you choose your routes optimally?

Constraints

$$2 \leq n \leq 500$$

$$1 \leq m \leq 1000$$

$$1 \leq a, b \leq n$$

Input

The first input line has two integers n and m : the number of rooms and teleporters. The rooms are numbered $1, 2, \dots, n$. After this, there are m lines describing the teleporters. Each line has two integers a and b : there is a teleporter from room a to room b . There are no two teleporters whose starting and ending room are the same.

Output

First print an integer k : the maximum number of days you can play the game. Then, print k route descriptions according to the example. You can print any valid solution.

▼ Logical Test Cases

Test Case 1

Test Case 2

INPUT (STDIN)

6 7
1 2
1 3
2 6
3 4
3 5
4 6
5 6

EXPECTED OUTPUT

2
4
1 3 5 6
3
1 2 6

INPUT (STDIN)

6 7
1 4
3 5
4 6
5 6
2 2
1 3
2 6

EXPECTED OUTPUT

2
4
1 3 5 6
3
1 4 6

▼ Mandatory Test Cases

Test Case 1

KEYWORD

int bfs(int n,int s,int t)

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

7

Test Case 2

TOKEN COUNT

920

Test Case 3

NLOC

110